

Hussein Ahmad

Full-Stack Web Developer & Machine Learning Specialist

Skills

Hussein Ahmad

Damascus, Syria

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<http://Paradome.pythonanywhere.com>

Programming Languages: Python, JavaScript

Frameworks & Libraries: Django, Django REST Framework, Machine Learning, Deep Learning, Deep learning frameworks, LangChain, Langgraph, DeepAgents, TensorFlow framework, Keras library, Flutter, JQuery, AJAX, HTMX

Databases: PostgreSQL, MySQL, Oracle, Chroma DB, PostGIS, Redis

DevOps & Tools: Docker, Nginx, RabbitMQ, MLOps, MLflow, AI Tools, Automated testing, Virtualization technologies, Continuous improvement, Agile methodologies, Continuous deployment, JIRA, Micro-Services, Linux environments, Source and version control (git, github, gitlab)

Other: Data visualization, API development, Task prioritization, Self motivation

Experience

LZ Helping Hands / Django and Devops engineer

Jan 2020 - PRESENT, Damascus

Developed ERP system using Django, PostgreSQL, RabbitMQ, Celery, and WebSockets for the back-end.

Integrated with a Flutter front-end.

Fully functional and scalable reports to adapt the institution needs.

Enhanced application performance by optimizing Python code and implementing Django best practices.

Developed custom Django apps to address unique business requirements and enhance system capabilities.

Deployed the ERP system for multi-devices offline with synchronization.

Duraa / Devops engineer

Jan 2023 - May 2025, Damascus

Used Django and Django REST Framework.

Implemented a full DevOps pipeline utilizing Docker, Nginx, Redis, and RabbitMQ for containerization, reverse proxying, and message queuing.

Implemented CI/CD pipelines with GitLab platform to automate builds, tests, and deployments of microservices.

Integrated PostGIS for geographic data, Celery for asynchronous task

management, and Firebase for additional services.

Dorar / AI engineer

Feb 2024 - Jul 2025, Remotely

Implemented scrapping websites, APIs and documents.

Enabled question-answering and semantic searching from private files.

Utilized Chroma DB for vector storage, LangChain for orchestration, and Jinaai for arabic embeddings.

Implemented Multi-Agent RAG Search architecture to enhance retrieval-augmented generation workflows, enabling collaborative agent-based querying and tool calling.

Eprocuments / AI engineer

mar 2025 - oct 2025, Freelancing

Project: AI-Powered Product Extraction & Classification Pipeline

Designed and built an end-to-end ETL pipeline to automate product data ingestion from unstructured sources (Web/Files) into structured database formats.

Integrated LLMs via LangChain to semantically analyze product descriptions and automate categorization, reducing manual data entry efforts by 90%.

Implemented Human-in-the-loop to review errors or low score.

Implemented asynchronous processing using Celery and RabbitMQ to handle high-volume data streams efficiently without system bottlenecks.

Ensured data quality by building validation scripts (Python/Pandas) before loading data into the final ERP/Database destination.

Other Contributions

Jan 2020 - PRESENT

- **Fine-tuned open-source large language models** using private datasets to fetch domain-specific insights.
- **Prepared custom datasets and fine-tuned language models using LoRA and Matryoshka embeddings**, optimizing parameter efficiency and embedding granularity for improved downstream task performance.
- **Applied transfer learning techniques** to adapt pre-trained models for custom NLP tasks.
- **Integrated ChromaDB and Weaviate** for efficient vector search and semantic retrieval, enhancing model-context relevance.
- **Used LangChain** for orchestration and pipeline automation.
- **Leveraged Ollama** for local LLM deployment.
- Developed Code-Agent based on HuggingFace to automate some workflows.
- Conducted evaluation and testing of fine-tuned models.

- Actively contributed to iterative improvements in model accuracy, contextual reasoning, and response relevance based on user feedback.
- Built a Django-based API to manage services, tickets, staff, and workers. Featured a notification system, Paired with a Flutter front-end for a complete user experience.
- Built Telegram bots integrated with Django for booking office and for tasks management.
- Developed a personal portfolio website, Created using Python with the Django framework, enhanced with JavaScript, Pusher for real-time features, and HTMX for dynamic front-end interactions.
- Worked independently to design, develop and test code, packages, and frameworks.

Education

Syrian Virtual University / Degree

Damascus

Bachelor in Information Technology Engineering specializing at Machine learning.

Coursera / Specialization

Enrolled and finished Machine Learning, Deep Learning and MLOps Specializations.